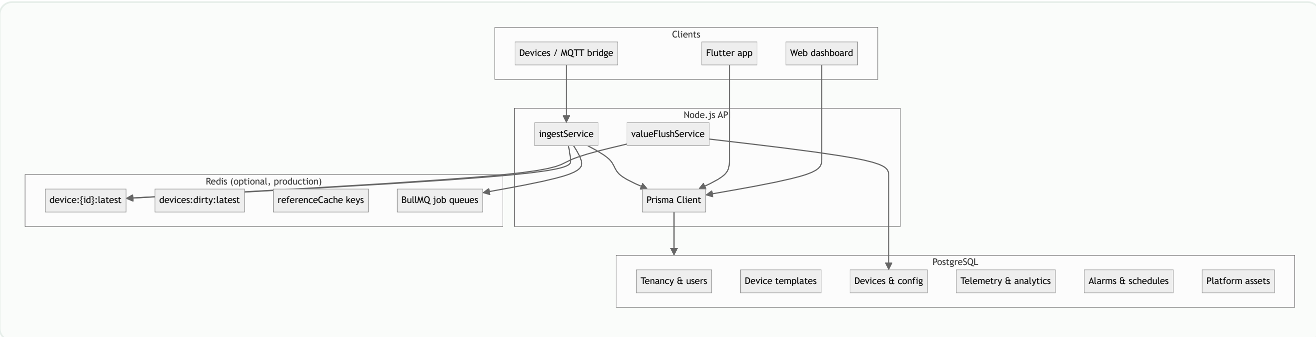
Smart AgriTech EMS Documentation

Database architecture

Complete reference for the Smart AgriTech EMS data layer: **PostgreSQL** (Prisma ORM) plus **Redis** for hot telemetry cache.

Source of truth: `ems/ems-backend/prisma/schema.prisma`
Migrations: `ems/ems-backend/prisma/migrations/`

Storage overview



Store	Engine	Purpose
Primary DB	PostgreSQL 14+	All persistent relational data
Hot cache	Redis 6+	Latest variable values, job queues, reference cache
ORM	Prisma 5+	Schema, migrations, type-safe queries

Multi-tenancy: Almost every operational table carries `organizationId` for row-level scoping. `SUPER_ADMIN` bypasses org filters in controllers; `ORG_ADMIN` and `USER` are restricted to their organization (and device assignments for users).

Domain model (high level)

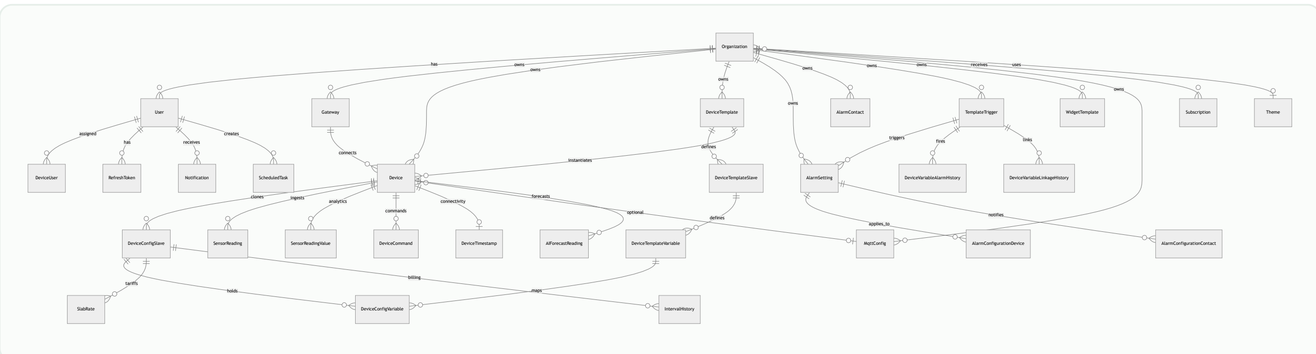


Table inventory (40 Prisma models)

#	Prisma model	SQL table	Domain
1	Organization	`organizations`	Tenancy
2	User	`users`	Auth
3	RefreshToken	`refresh_tokens`	Auth
4	PasswordResetCode	`password_reset_codes`	Auth
5	Gateway	`gateways`	IoT
6	DeviceTemplate	`device_templates`	IoT template
7	DeviceTemplateSlave	`device_template_slaves`	IoT template
8	DeviceTemplateVariable	`device_template_variables`	IoT template
9	Device	`devices`	IoT runtime
10	DeviceConfigSlave	`device_config_slaves`	IoT runtime
11	DeviceConfigVariable	`device_config_variables`	IoT runtime
12	DeviceConfigVariableLog	`device_config_variable_logs`	IoT audit
13	DeviceUser	`device_users`	Access control
14	DeviceCommand	`device_commands`	IoT commands
15	MqttConfig	`mqtt_configs`	IoT connectivity
16	SensorReading	`sensor_readings`	Telemetry
17	SensorReadingValue	`sensor_reading_values`	Telemetry analytics
18	DeviceTimestamp	`device_timestamps`	Connectivity
19	TemplateTrigger	`template_triggers`	Alarms
20	AlarmSetting	`alarm_settings`	Alarms
21	AlarmConfigurationDevice	`alarm_configuration_devices`	Alarms M:N
22	AlarmConfigurationContact	`alarm_configuration_contacts`	Alarms M:N
23	AlarmContact	`alarm_contacts`	Alarms
24	AlarmHistoryNotification	`alarm_history_notifications`	Alarms
25	DeviceVariableAlarmHistory	`device_variable_alarm_histories`	Alarms
26	DeviceVariableLinkageHistory	`device_variable_linkage_histories`	Alarms
27	ScheduledTask	`scheduled_tasks`	Automation
28	ScheduleExecutionLog	`schedule_execution_logs`	Automation
29	SlabRate	`slab_rates`	Billing
30	IntervalHistory	`interval_histories`	Billing
31	AIForecastReading	`ai_forecast_readings`	Analytics
32	Notification	`notifications`	User inbox

#	Prisma model	SQL table	Domain
33	Icon	`icons`	UI assets
34	Product	`products`	Catalog
35	Theme	`themes`	UI theming
36	WidgetTemplate	`widget_templates`	Dashboard widgets
37	SystemSetting	`system_settings`	Platform config
38	ListType	`list_types`	Lookup lists
39	ListTypeItem	`list_type_items`	Lookup lists
40	Subscription	`subscriptions`	Lead capture

> **Legacy note:** Initial migration also created a `predictions` table. The application uses `ai_forecast_readings` (Prisma `AIForecastReading`). The old `predictions` table may exist on databases created from the init migration only — it is not in the current Prisma schema.

Enumerations (PostgreSQL ENUM types)

Enum	Values	Used by
`OrgStatus`	`ACTIVE`, `INACTIVE`	organizations
`Role`	`SUPER_ADMIN`, `ORG_ADMIN`, `USER`	users
`UserStatus`	`ACTIVE`, `INACTIVE`, `DELETED`	users
`DeviceStatus`	`ONLINE`, `OFFLINE`	gateways, devices
`DataType`	`FLOAT`, `INTEGER`, `BOOLEAN`, `STRING`	device_template_variables
`SwitchState`	`ON`, `OFF`	devices
`CommandStatus`	`PENDING`, `ACKNOWLEDGED`, `FAILED`, `TIMEOUT`	device_commands
`LogSource`	`INGEST`, `MANUAL`, `SCHEDULE`, `AUTOMATION`	device_config_variable_logs
`Operator`	`GT`, `LT`, `EQ`, `GTE`, `LTE`	template_triggers
`Priority`	`LOW`, `MEDIUM`, `HIGH`	template_triggers
`AlarmStatus`	`ACTIVE`, `INACTIVE`	alarm_settings
`NotificationStatus`	`SENT`, `FAILED`	alarm_history_notifications
`AlarmState`	`ACTIVE`, `RESOLVED`	device_variable_alarm_histories
`ProcessState`	`UNPROCESSED`, `PROCESSED`	device_variable_alarm_histories
`TaskAction`	`ON`, `OFF`	scheduled_tasks
`RepeatType`	`DAILY`, `WEEKLY`, `ONCE`	scheduled_tasks
`TaskStatus`	`ACTIVE`, `INACTIVE`	scheduled_tasks
`ExecutionResult`	`SUCCESS`, `FAILED`	schedule_execution_logs

Enum	Values	Used by
`Horizon`	`TEN_MIN` , `FIVE_HR` , `SEVEN_DAY` , `CUSTOM`	ai_forecast_readings
`IconStatus`	`ACTIVE` , `INACTIVE`	icons
`ProductStatus`	`ACTIVE` , `INACTIVE`	products
`ThemeStatus`	`ACTIVE` , `INACTIVE`	themes
`WidgetType`	`BAR` , `LINE` , `AREA` , `GAUGE` , `VALUE_CARD` , `PIE`	widget_templates
`SubscriptionStatus`	`NEW` , `CONTACTED` , `CLOSED`	subscriptions

Keys & indexes summary

Primary keys

All tables use `id TEXT PRIMARY KEY` with UUID generated in application (`@default(uuid())`).

Unique constraints

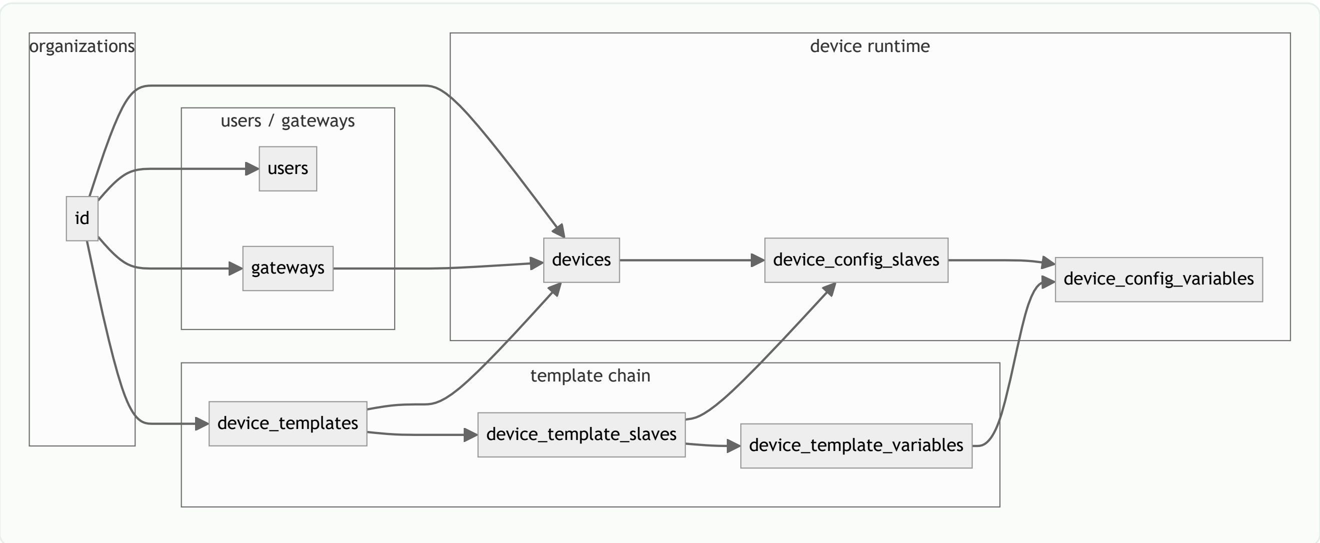
Table	Constraint	Columns
users	`users_email_key`	`email`
gateways	`gateways_serialNumber_key`	`serialNumber`
device_template_variables	`device_template_variables_templateSlaveId_name_key`	`(templateSlaveId, name)`
devices	`devices_mqttConfigId_key`	`mqttConfigId`
device_config_variables	`device_config_variables_deviceId_deviceConfigSlaveId_name_key`	`(deviceId, deviceConfigSlaveId, name)`
device_users	`device_users_deviceId_userId_key`	`(deviceId, userId)`
device_timestamps	`device_timestamps_deviceId_key`	`deviceId`
refresh_tokens	`refresh_tokens_tokenHash_key`	`tokenHash`
alarm_configuration_devices	`alarm_configuration_devices_alarmSettingId_deviceId_key`	`(alarmSettingId, deviceId)`
alarm_configuration_contacts	`alarm_configuration_contacts_alarmSettingId_alarmContactId_key`	`(alarmSettingId, alarmContactId)`
system_settings	`system_settings_key_key`	`key`
list_types	`list_types_name_key`	`name`

Performance indexes

Table	Index	Columns	Query pattern
refresh_tokens	`refresh_tokens_userId_idx`	`userId`	Logout all sessions

Table	Index	Columns	Query pattern
device_config_variable_logs	composite	`(deviceConfigVariableId, changedAt)`	Variable change history
device_config_variable_logs	composite	`(deviceId, changedAt)`	Device audit trail
device_users	composite	`(userId, deviceId)`	User device list
sensor_readings	composite	`(deviceId, deviceConfigSlaveId, timestamp)`	Raw history by slave
sensor_readings	composite	`(deviceId, timestamp)`	Device time series
sensor_readings	composite	`(organizationId, timestamp)`	Org-wide queries
sensor_reading_values	composite	`(deviceId, variableName, timestamp)`	Analytics charts
sensor_reading_values	composite	`(deviceId, timestamp)`	Dashboard aggregates
sensor_reading_values	composite	`(organizationId, timestamp)`	Org analytics
sensor_reading_values	single	`sensorReadingId`	Join raw → normalized
device_commands	composite	`(deviceId, status)`	Pending commands
device_commands	composite	`(organizationId, requestedAt)`	Admin command log
device_variable_alarm_histories	composite	`(deviceId, alarmTime)`	Alarm history
device_variable_linkage_histories	composite	`(deviceId, firedAt)`	Linkage audit
schedule_execution_logs	composite	`(scheduleTaskId, executedAt)`	Task run history
notifications	composite	`(userId, createdAt)`	User inbox

Foreign key linkage map

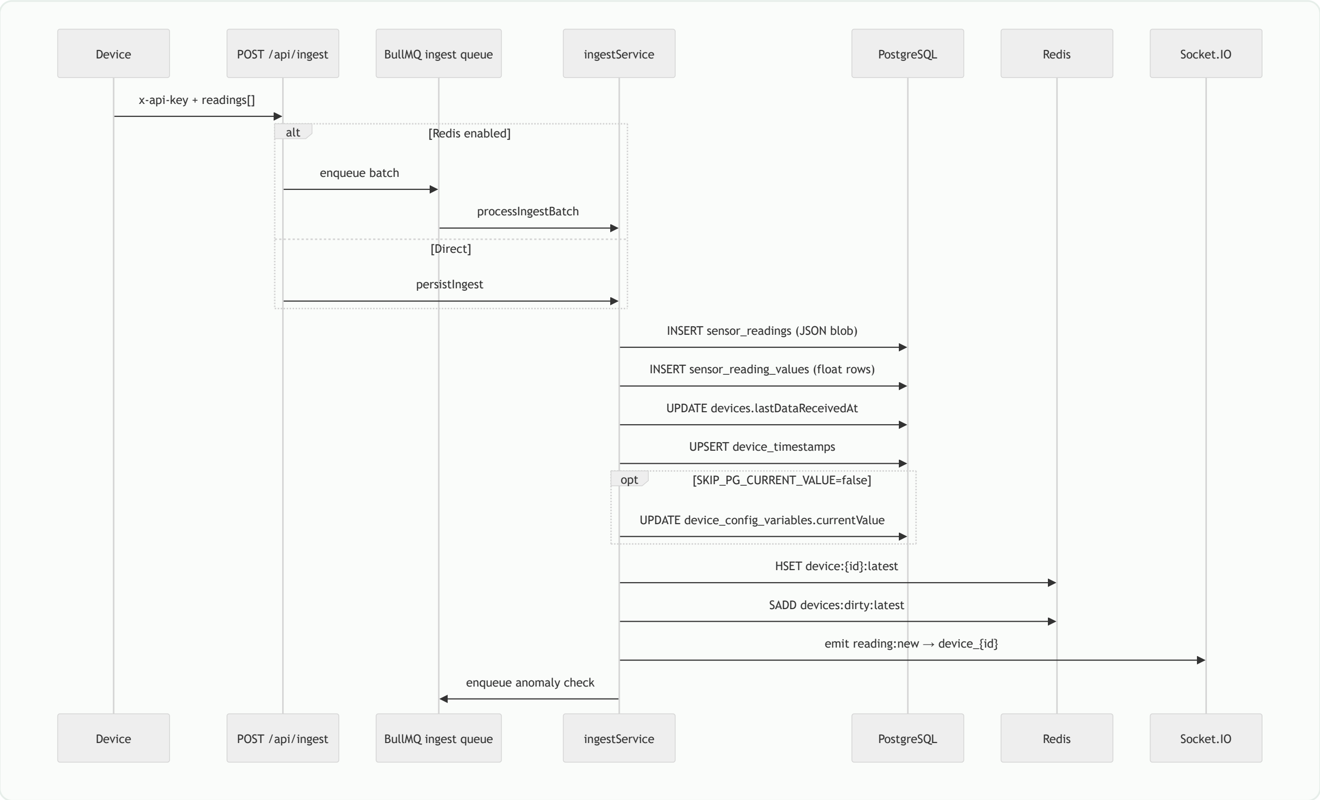


FK delete behavior

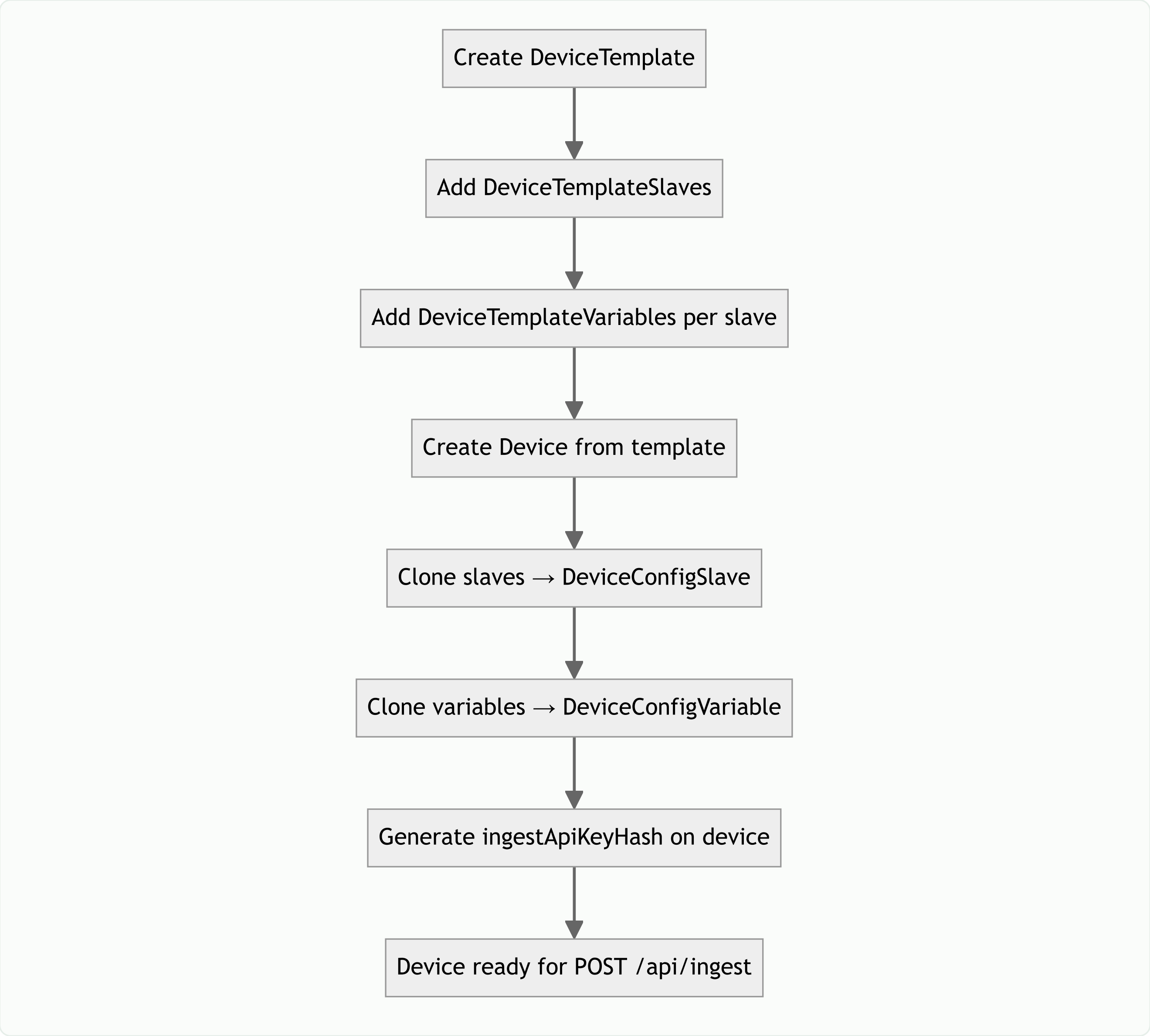
Parent → Child	ON DELETE
Organization → Gateway, DeviceTemplate, Device, …	`RESTRICT` (cannot delete org with children)
Organization → User, WidgetTemplate, Subscription	`SET NULL`
Device → SensorReading, DeviceConfigSlave, DeviceUser, …	`CASCADE`
DeviceTemplate → DeviceTemplateSlave	`CASCADE`
DeviceTemplateSlave → DeviceTemplateVariable	`CASCADE`
DeviceConfigSlave → SensorReading	`SET NULL` on slave delete
AlarmSetting → AlarmConfigurationDevice	`CASCADE`
TemplateTrigger → AlarmSetting	`SET NULL`

Data flow diagrams

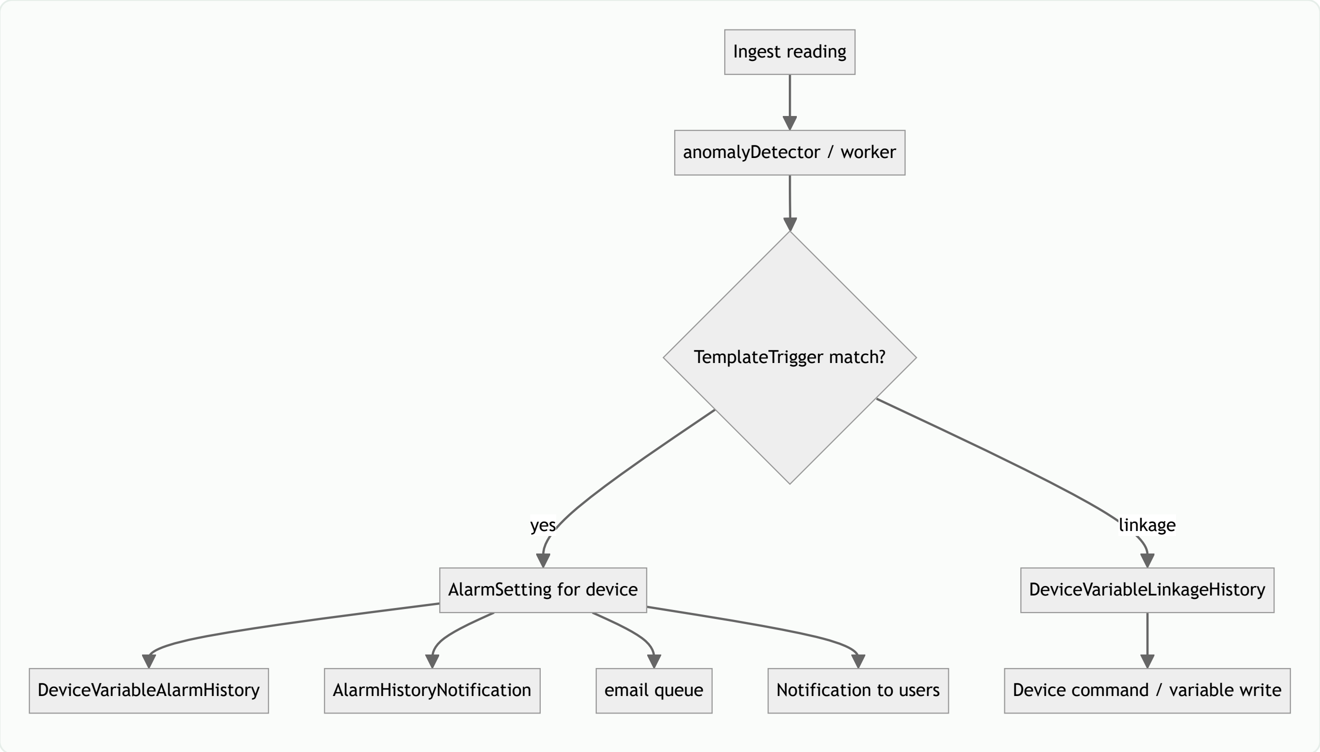
Ingest → PostgreSQL + Redis



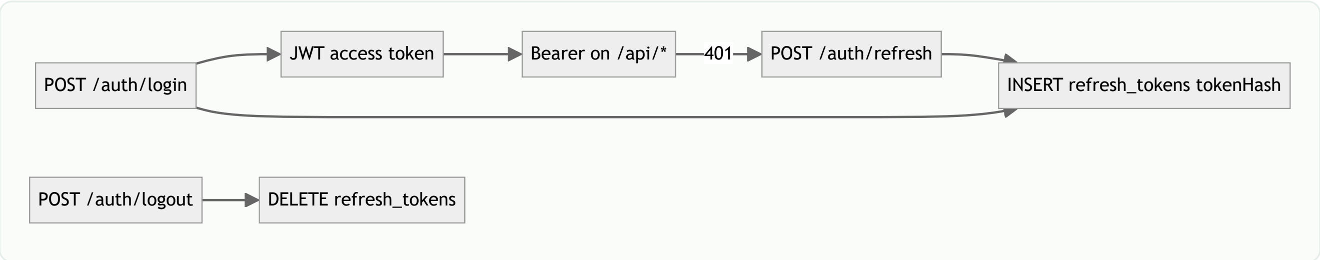
Template → device instantiation



Alarm evaluation chain



Auth token flow



Redis key catalog

Redis is **not** defined in Prisma — these keys are used at runtime.

Key pattern	Type	TTL	Written by	Read by
`device:{deviceId}:latest`	Hash (field = variableName, value = string)	3600s	ingestService	sensorDataController `/latest`, valueFlushService
`devices:dirty:latest`	Set of deviceIds	—	valueFlushService.markDirty	valueFlushService flush loop
BullMQ `ingest`	Queue	—	ingest route	ingest worker
BullMQ `anomaly`	Queue	—	ingestService	anomaly worker
BullMQ `email`	Queue	—	alarm notifier	email worker
BullMQ `device-delete`	Queue	—	device delete	purge worker
`ref:*` (referenceCache)	String/Hash	configurable	referenceCache utils	controllers on cache miss

Flush cycle: When `SKIP_PG_CURRENT_VALUE=true` (default with Redis), `currentValue` in Postgres is updated periodically from Redis hash via `valueFlushService` (`VALUE_FLUSH_MS` , default 60s).

Table reference (all columns)

1. organizations

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	Primary key
name	TEXT	NO	—	Organization display name
description	TEXT	YES	—	Optional description
status	OrgStatus	NO	ACTIVE	ACTIVE / INACTIVE
themeId	TEXT	YES	—	FK → themes.id
logoUrl	TEXT	YES	—	Logo image URL
createdAt	TIMESTAMP	NO	now()	Created
updatedAt	TIMESTAMP	NO	—	Auto-updated

Relations: users, devices, gateways, deviceTemplates, alarmConfigurations, alarmContacts, templateTriggers, widgetTemplates, mqttConfigs, subscriptions

---

2. users

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
fullName	TEXT	NO	—	Display name
email	TEXT	NO	—	**Unique** login
passwordHash	TEXT	NO	—	bcrypt hash
phone	TEXT	YES	—	Optional phone
role	Role	NO	USER	SUPER_ADMIN / ORG_ADMIN / USER
organizationId	TEXT	YES	—	FK → organizations (null for super admin)
status	UserStatus	NO	ACTIVE	ACTIVE / INACTIVE / DELETED
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

---

3. refresh_tokens

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK

Column	Type	Null	Default	Description
userId	TEXT	NO	—	FK → users (CASCADE)
tokenHash	TEXT	NO	—	**Unique** SHA-256 of refresh token
expiresAt	TIMESTAMP	NO	—	Expiry
createdAt	TIMESTAMP	NO	now()	—

4. password_reset_codes

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
userId	TEXT	NO	—	FK → users (CASCADE)
code	TEXT	NO	—	Reset code
expiresAt	TIMESTAMP	NO	—	Expiry
used	BOOLEAN	NO	false	One-time use flag
createdAt	TIMESTAMP	NO	now()	—

5. gateways

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
name	TEXT	NO	—	Gateway name
serialNumber	TEXT	NO	—	**Unique** hardware ID
model	TEXT	YES	—	Model string
status	DeviceStatus	NO	OFFLINE	ONLINE / OFFLINE
organizationId	TEXT	NO	—	FK → organizations
lastSeenAt	TIMESTAMP	YES	—	Last heartbeat
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

6. device_templates

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
name	TEXT	NO	—	Template name
organizationId	TEXT	NO	—	FK → organizations
acquisitionMethod	TEXT	YES	—	e.g. Modbus, MQTT

Column	Type	Null	Default	Description
totalSlaves	INT	NO	0	Denormalized count
totalVariables	INT	NO	0	Denormalized count
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

---

7. device_template_slaves

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
templateId	TEXT	NO	—	FK → device_templates (CASCADE)
organizationId	TEXT	NO	—	Org scope
name	TEXT	NO	—	Slave name (e.g. Meter 1)
description	TEXT	YES	—	—
isDefault	BOOLEAN	NO	false	Default slave for UI
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

---

8. device_template_variables

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
templateSlaveId	TEXT	NO	—	FK → device_template_slaves
templateId	TEXT	NO	—	Denormalized template ref
organizationId	TEXT	NO	—	Org scope
name	TEXT	NO	—	Variable key in ingest JSON
displayName	TEXT	YES	—	UI label
unit	TEXT	YES	—	e.g. kW, V, °C
registerAddress	TEXT	YES	—	Modbus register
iconId	TEXT	YES	—	FK → icons
dataType	DataType	NO	FLOAT	FLOAT / INTEGER / BOOLEAN / STRING
isActive	BOOLEAN	NO	true	Soft disable
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

Unique: `(templateSlaveId, name)`

---

9. devices

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK; also used in ingest payload
name	TEXT	NO	—	Device display name
gatewayId	TEXT	YES	—	FK → gateways
organizationId	TEXT	NO	—	FK → organizations
templateId	TEXT	NO	—	FK → device_templates
switchState	SwitchState	NO	OFF	ON / OFF (scheduler)
status	DeviceStatus	NO	OFFLINE	Connectivity
mqttConfigId	TEXT	YES	—	**Unique** FK → mqtt_configs (1:1)
lastDataReceivedAt	TIMESTAMP	YES	—	Last ingest time
ingestApiKeyHash	TEXT	YES	—	SHA-256 of per-device ingest key
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

Ingest auth: `ingestApiKeyHash` compared against `x-api-key` header (or global `INGEST_API_KEY` env).

---

10. device_config_slaves

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK; used as `slaveId` in ingest
deviceId	TEXT	NO	—	FK → devices (CASCADE)
templateSlaveId	TEXT	NO	—	FK → device_template_slaves
organizationId	TEXT	NO	—	Org scope
name	TEXT	NO	—	Runtime slave name
description	TEXT	YES	—	—
isDefault	BOOLEAN	NO	false	—
isActive	BOOLEAN	NO	true	—
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

---

11. device_config_variables

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
deviceId	TEXT	NO	—	FK → devices (CASCADE)
deviceConfigSlaveId	TEXT	NO	—	FK → device_config_slaves

Column	Type	Null	Default	Description
templateVariableId	TEXT	NO	—	FK → device_template_variables
organizationId	TEXT	NO	—	Org scope
name	TEXT	NO	—	Must match ingest `variableName`
displayName	TEXT	YES	—	UI label
unit	TEXT	YES	—	—
currentValue	TEXT	YES	—	Latest value (PG or flushed from Redis)
lastUpdatedAt	TIMESTAMP	YES	—	Last value change
isActive	BOOLEAN	NO	true	—
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

Unique: `(deviceId, deviceConfigSlaveId, name)`

---

12. device_config_variable_logs

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
deviceConfigVariableId	TEXT	NO	—	FK → device_config_variables
deviceId	TEXT	NO	—	FK → devices
organizationId	TEXT	NO	—	Org scope
previousValue	TEXT	YES	—	Before change
newValue	TEXT	YES	—	After change
source	LogSource	NO	INGEST	INGEST / MANUAL / SCHEDULE / AUTOMATION
changedAt	TIMESTAMP	NO	now()	—

---

13. device_users

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
deviceId	TEXT	NO	—	FK → devices (CASCADE)
userId	TEXT	NO	—	FK → users (CASCADE)
organizationId	TEXT	NO	—	Org scope
assignedAt	TIMESTAMP	NO	now()	—
assignedBy	TEXT	YES	—	Admin user id

Unique: `(deviceId, userId)` — controls USER role device access.

---

14. device_commands

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
deviceId	TEXT	NO	—	FK → devices (CASCADE)
organizationId	TEXT	NO	—	Org scope
action	TEXT	NO	—	Command payload
status	CommandStatus	NO	PENDING	PENDING / ACKNOWLEDGED / FAILED / TIMEOUT
requestedBy	TEXT	YES	—	User id
requestedAt	TIMESTAMP	NO	now()	—
acknowledgedAt	TIMESTAMP	YES	—	Device ACK time
failedReason	TEXT	YES	—	Error detail

---

15. mqtt_configs

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
organizationId	TEXT	NO	—	FK → organizations
brokerUrl	TEXT	YES	—	MQTT broker host
port	INT	NO	1883	Broker port
username	TEXT	YES	—	—
passwordEncrypted	TEXT	YES	—	Encrypted password
topic	TEXT	YES	—	Subscribe topic
isActive	BOOLEAN	NO	true	—
createdBy	TEXT	YES	—	FK → users
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

Relation: One optional `Device` per config via `devices.mqttConfigId`.

---

16. sensor_readings (raw ingest log)

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
deviceId	TEXT	NO	—	FK → devices (CASCADE)
deviceConfigSlaveId	TEXT	YES	—	FK → device_config_slaves (SET NULL)
organizationId	TEXT	NO	—	Org scope
timestamp	TIMESTAMP	NO	now()	Ingest time

Column	Type	Null	Default	Description
readings	JSON	NO	—	Full batch `{ variableName, value }[]`

Usage: Sensor History page (`GET /sensor-data/readings`), audit, replay.

---

17. sensor_reading_values (normalized analytics)

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
sensorReadingId	TEXT	NO	—	Parent reading id (no FK in schema)
deviceId	TEXT	NO	—	Denormalized device
deviceConfigSlaveId	TEXT	YES	—	Optional slave
organizationId	TEXT	NO	—	Org scope
variableName	TEXT	NO	—	Variable key
value	FLOAT	NO	—	Numeric value
timestamp	TIMESTAMP	NO	—	Reading time

Usage: Charts, aggregates (`GET /sensor-data/history` , `/aggregate` , `/dashboard-summary`), AI analytics.

> Denormalized by design for fast time-series queries without JSON parsing.

---

18. device_timestamps

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
deviceId	TEXT	NO	—	**Unique** FK → devices
organizationId	TEXT	NO	—	Org scope
lastActiveAt	TIMESTAMP	NO	now()	Last ingest / activity

---

19. template_triggers

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
deviceTemplateId	TEXT	NO	—	FK → device_templates (CASCADE)
organizationId	TEXT	NO	—	FK → organizations
name	TEXT	NO	—	Trigger name
templateVariableId	TEXT	NO	—	FK → device_template_variables (watched)

Column	Type	Null	Default	Description
operator	Operator	NO	—	GT / LT / EQ / GTE / LTE
threshold	FLOAT	NO	—	Compare value
anomalyType	TEXT	NO	—	Classification label
priority	Priority	NO	MEDIUM	LOW / MEDIUM / HIGH
linkageVariableId	TEXT	YES	—	FK → device_template_variables (action target)
linkageAction	TEXT	YES	—	e.g. SET, TOGGLE
linkageValue	TEXT	YES	—	Value to write
isActive	BOOLEAN	NO	true	—
createdBy	TEXT	YES	—	FK → users
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

---

20. alarm_settings

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
name	TEXT	NO	—	Alarm config name
organizationId	TEXT	NO	—	FK → organizations
templateTriggerId	TEXT	YES	—	FK → template_triggers
pushType	TEXT	YES	—	Notification type
pushBody	TEXT	YES	—	Message template
pushMethod	TEXT	YES	—	email / sms / push
pushingMechanism	TEXT	YES	—	Delivery mechanism
status	AlarmStatus	NO	ACTIVE	ACTIVE / INACTIVE
createdBy	TEXT	YES	—	FK → users
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

M:N: devices via `alarm_configuration_devices`, contacts via `alarm_configuration_contacts`.

---

21. alarm_configuration_devices

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
alarmSettingId	TEXT	NO	—	FK → alarm_settings (CASCADE)
deviceId	TEXT	NO	—	FK → devices (CASCADE)

Unique: `(alarmSettingId, deviceId)`

---

22. alarm_configuration_contacts

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
alarmSettingId	TEXT	NO	—	FK → alarm_settings (CASCADE)
alarmContactId	TEXT	NO	—	FK → alarm_contacts (CASCADE)

Unique: `(alarmSettingId, alarmContactId)`

---

23. alarm_contacts

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
name	TEXT	NO	—	Contact name
organizationId	TEXT	NO	—	FK → organizations
mobile	TEXT	YES	—	Phone
email	TEXT	YES	—	Email
whatsapp	TEXT	YES	—	WhatsApp
remark	TEXT	YES	—	Notes
createdBy	TEXT	YES	—	FK → users
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

---

24. alarm_history_notifications

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
alarmSettingId	TEXT	YES	—	FK → alarm_settings
organizationId	TEXT	NO	—	Org scope
deviceId	TEXT	YES	—	FK → devices (SET NULL)
message	TEXT	YES	—	Sent message
pushType	TEXT	YES	—	Channel
sentTo	TEXT	YES	—	Recipient
sentAt	TIMESTAMP	NO	now()	—
status	NotificationStatus	NO	SENT	SENT / FAILED

---

25. device_variable_alarm_histories

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
alarmSettingId	TEXT	YES	—	FK → alarm_settings
templateTriggerId	TEXT	YES	—	FK → template_triggers
deviceId	TEXT	NO	—	FK → devices (CASCADE)
organizationId	TEXT	NO	—	Org scope
variableName	TEXT	NO	—	Variable that triggered
triggerName	TEXT	YES	—	Snapshot of trigger name
triggerType	TEXT	YES	—	Anomaly type
slaveName	TEXT	YES	—	Slave label
currentValue	FLOAT	YES	—	Value at alarm time
triggeringCondition	TEXT	YES	—	Human-readable condition
alarmTime	TIMESTAMP	NO	now()	—
alarmState	AlarmState	NO	ACTIVE	ACTIVE / RESOLVED
processState	ProcessState	NO	UNPROCESSED	UNPROCESSED / PROCESSED

26. device_variable_linkage_histories

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
deviceId	TEXT	NO	—	FK → devices (CASCADE)
organizationId	TEXT	NO	—	Org scope
templateTriggerId	TEXT	YES	—	FK → template_triggers
triggerName	TEXT	YES	—	—
watchedVariableName	TEXT	YES	—	Input variable
watchedVariableValue	FLOAT	YES	—	Input value
linkedVariableName	TEXT	YES	—	Output variable
actionTaken	TEXT	YES	—	Action performed
firedAt	TIMESTAMP	NO	now()	—

27. scheduled_tasks

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
organizationId	TEXT	NO	—	Org scope

Column	Type	Null	Default	Description
createdBy	TEXT	YES	—	FK → users
deviceId	TEXT	NO	—	FK → devices (CASCADE)
deviceConfigSlaveId	TEXT	YES	—	FK → device_config_slaves (SET NULL)
deviceConfigVariableId	TEXT	YES	—	FK → device_config_variables
variableName	TEXT	NO	—	Target variable
action	TaskAction	NO	—	ON / OFF
scheduledTime	TEXT	NO	—	Time expression (cron-like)
repeatType	RepeatType	NO	DAILY	DAILY / WEEKLY / ONCE
daysOfWeek	INT[]	NO	—	PostgreSQL integer array
status	TaskStatus	NO	ACTIVE	ACTIVE / INACTIVE
nextRunAt	TIMESTAMP	YES	—	Next execution
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

---

28. schedule_execution_logs

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
scheduleTaskId	TEXT	NO	—	FK → scheduled_tasks (CASCADE)
deviceId	TEXT	NO	—	FK → devices (CASCADE)
organizationId	TEXT	NO	—	Org scope
executedAt	TIMESTAMP	NO	now()	—
action	TEXT	YES	—	Action executed
variableName	TEXT	YES	—	—
result	ExecutionResult	NO	SUCCESS	SUCCESS / FAILED
errorMessage	TEXT	YES	—	Failure reason

---

29. slab_rates

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
organizationId	TEXT	NO	—	Org scope
deviceConfigSlaveId	TEXT	NO	—	FK → device_config_slaves (CASCADE)
unitFrom	FLOAT	NO	—	Tier start (kWh)
unitTo	FLOAT	NO	—	Tier end

Column	Type	Null	Default	Description
rate	FLOAT	NO	—	Price per unit
onPeakRate	FLOAT	YES	—	Peak tariff
offPeakRate	FLOAT	YES	—	Off-peak tariff
createdBy	TEXT	YES	—	FK → users
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

30. interval_histories

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
organizationId	TEXT	NO	—	Org scope
deviceId	TEXT	YES	—	FK → devices (SET NULL)
deviceConfigSlaveId	TEXT	NO	—	FK → device_config_slaves (CASCADE)
templateVariableId	TEXT	YES	—	FK → device_template_variables
templateSlaveId	TEXT	YES	—	FK → device_template_slaves
variableName	TEXT	NO	—	Metered variable
slaveName	TEXT	YES	—	Display
totalUnit	FLOAT	NO	0	Consumption in period
tariff	FLOAT	NO	0	Applied rate
startDate	TIMESTAMP	NO	—	Billing period start
endDate	TIMESTAMP	NO	—	Billing period end
computedAt	TIMESTAMP	NO	now()	Calculation time

31. ai_forecast_readings

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
deviceId	TEXT	NO	—	FK → devices (CASCADE)
organizationId	TEXT	NO	—	Org scope
templateVariableId	TEXT	YES	—	FK → device_template_variables
templateSlaveId	TEXT	YES	—	FK → device_template_slaves
variableName	TEXT	NO	—	Forecast target
horizon	Horizon	NO	—	TEN_MIN / FIVE_HR / SEVEN_DAY / CUSTOM
predictions	JSON	NO	—	`[{ timestamp, value }]` array

Column	Type	Null	Default	Description
generatedAt	TIMESTAMP	NO	now()	—

32. notifications

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
userId	TEXT	NO	—	FK → users (CASCADE)
organizationId	TEXT	NO	—	Org scope
triggerName	TEXT	YES	—	Alarm trigger label
deviceName	TEXT	YES	—	Device label
description	TEXT	YES	—	Message body
read	BOOLEAN	NO	false	Read flag
createdAt	TIMESTAMP	NO	now()	—

33. icons

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
name	TEXT	NO	—	Icon name
imageUrl	TEXT	NO	—	Image URL (Cloudinary)
status	IconStatus	NO	ACTIVE	ACTIVE / INACTIVE
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

34. products

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
name	TEXT	NO	—	Product name
price	FLOAT	YES	—	Price
imageUrl	TEXT	YES	—	Image
description	TEXT	YES	—	—
status	ProductStatus	NO	ACTIVE	ACTIVE / INACTIVE
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

35. themes

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
name	TEXT	NO	—	Theme name
headerFontColor	TEXT	YES	—	CSS color
headerBgColor	TEXT	YES	—	CSS color
bodyFontColor	TEXT	YES	—	CSS color
bodyBgColor	TEXT	YES	—	CSS color
fontSize	TEXT	YES	—	Base font size
status	ThemeStatus	NO	ACTIVE	—
createdBy	TEXT	YES	—	FK → users
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

36. widget_templates

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
organizationId	TEXT	YES	—	FK → organizations
name	TEXT	NO	—	Widget name
iconId	TEXT	YES	—	FK → icons
themeId	TEXT	YES	—	FK → themes
widgetType	WidgetType	NO	VALUE_CARD	BAR / LINE / AREA / GAUGE / VALUE_CARD / PIE
variableName	TEXT	YES	—	Bound variable
displayName	TEXT	YES	—	Label
unit	TEXT	YES	—	—
position	INT	NO	0	Dashboard sort order
isActive	BOOLEAN	NO	true	—
createdBy	TEXT	YES	—	FK → users
createdAt	TIMESTAMP	NO	now()	—
updatedAt	TIMESTAMP	NO	—	—

37. system_settings

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK

Column	Type	Null	Default	Description
key	TEXT	NO	—	**Unique** setting key
type	TEXT	NO	—	string / number / boolean
value	TEXT	YES	—	Serialized value
description	TEXT	YES	—	Admin description
updatedAt	TIMESTAMP	NO	—	—

---

38. list_types / 39. list_type_items

list_types

Column	Type	Description
id	TEXT PK	—
name	TEXT **unique**	List category name
description	TEXT	—
isActive	BOOLEAN	—

list_type_items

Column	Type	Description
id	TEXT PK	—
listTypeId	TEXT FK	→ list_types (CASCADE)
name	TEXT	Item label
description	TEXT	—
isActive	BOOLEAN	—

---

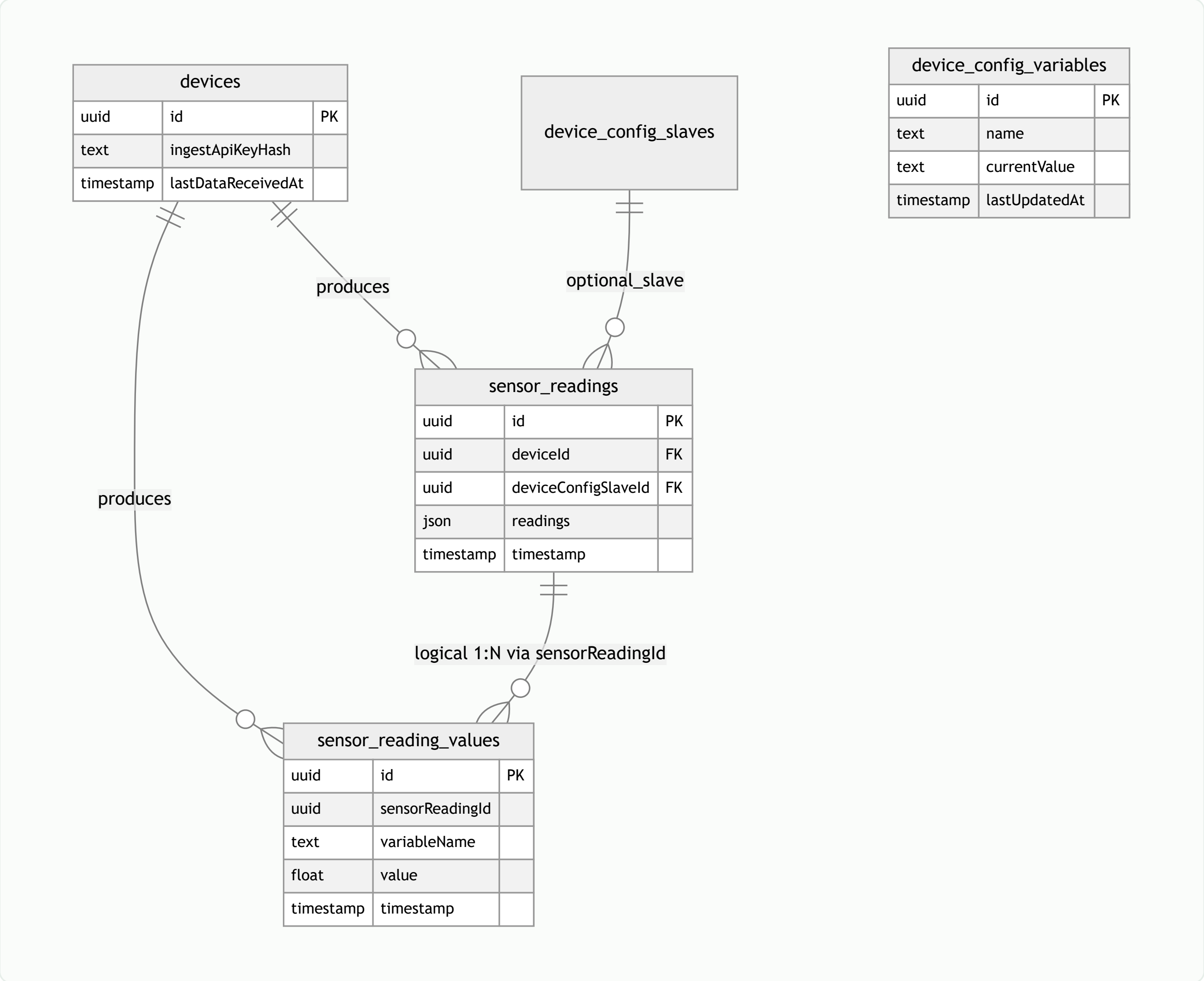
40. subscriptions

Column	Type	Null	Default	Description
id	TEXT	NO	uuid	PK
organizationId	TEXT	YES	—	FK → organizations
name	TEXT	NO	—	Requester name
email	TEXT	NO	—	Contact email
phone	TEXT	YES	—	—
description	TEXT	YES	—	Message
status	SubscriptionStatus	NO	NEW	NEW / CONTACTED / CLOSED
submittedAt	TIMESTAMP	NO	now()	—

Feature → table mapping

Feature	Primary tables
Login / JWT	users, refresh_tokens
Password reset	password_reset_codes
Org management	organizations, themes
User management	users, device_users
Gateway registry	gateways
Device onboarding	devices, device_templates, device_config_*
Ingest telemetry	sensor_readings, sensor_reading_values, device_config_variables, device_timestamps
Live dashboard	device_config_variables + Redis `device:*:latest`
Sensor history	sensor_readings, sensor_reading_values
Charts / aggregates	sensor_reading_values
Device commands	device_commands
Alarms	template_triggers, alarm_settings, alarm_configuration_*, device_variable_alarm_histories
Linkage automation	template_triggers, device_variable_linkage_histories
Notifications	notifications, alarm_history_notifications
Schedules	scheduled_tasks, schedule_execution_logs
Billing	slab_rates, interval_histories
AI analytics	sensor_reading_values, ai_forecast_readings
Dashboard widgets	widget_templates, icons, themes
MQTT	mqtt_configs, devices
Platform catalog	products, icons, system_settings, list_types
Subscription leads	subscriptions

Entity relationship (telemetry focus)



Security-sensitive columns

Table	Column	Storage	Notes
users	passwordHash	bcrypt	Never returned in API
devices	ingestApiKeyHash	SHA-256	Plain key shown once on create
refresh_tokens	tokenHash	SHA-256	Raw token only in HTTP response
mqtt_configs	passwordEncrypted	Encrypted	Broker credentials
password_reset_codes	code	Plain	Short-lived, single use

Maintenance & operations

Task	Command / approach
Apply migrations	`npx prisma migrate deploy`

Task	Command / approach
Regenerate client	`npx prisma generate`
Seed demo data	`npm run seed`
Bulk delete sensor data	`DELETE /api/sensor-data` (SUPER_ADMIN)
Device purge	BullMQ `device-delete` queue cascades related rows
Backup	`pg_dump` of CapRover `iotpostgres` volume
Read replica	Set `DATABASE_READ_URL` for read-heavy queries

Related documents

- Architecture
- Backend API
- System flows
- Deployment
- Prisma schema: `ems/ems-backend/prisma/schema.prisma`